

Machine Learning - 70 Interview Questions with Answers

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Overview: Types of Machine Learning

1. Supervised Learning

Definition: Learning from labeled data (input-output pairs)

Types:

- **Classification:** Predicting discrete labels
 - Binary: Spam/Not Spam, Yes/No
 - Multi-class: Digit recognition (0-9), Image classification
 - Multi-label: Tag prediction (multiple tags per item)
- **Regression:** Predicting continuous values
 - Linear regression, polynomial regression
 - House price prediction, stock price forecasting

Common Algorithms:

- Linear/Logistic Regression
- Decision Trees, Random Forest
- SVM, Naive Bayes, KNN
- Neural Networks
- Gradient Boosting (XGBoost, LightGBM, CatBoost)

Use Cases:

- Email spam detection
 - Fraud detection
 - Medical diagnosis
 - Sales forecasting
 - Image/speech recognition
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2. Unsupervised Learning

Definition: Learning from unlabeled data to find patterns

Types:

- **Clustering:** Grouping similar data points
 - K-Means, DBSCAN, Hierarchical Clustering
 - Gaussian Mixture Models (GMM)
- **Dimensionality Reduction:** Reducing number of features
 - PCA (Principal Component Analysis)
 - t-SNE, UMAP (visualization)
 - Autoencoders
- **Anomaly Detection:** Finding outliers
 - Isolation Forest
 - One-Class SVM
 - LOF (Local Outlier Factor)
- **Association Rule Learning:** Finding relationships
 - Apriori algorithm
 - FP-Growth
 - Market basket analysis

Use Cases:

- Customer segmentation
 - Recommendation systems (user/item clustering)
 - Data compression
 - Feature extraction
 - Anomaly detection (fraud, network intrusion)
 - Market basket analysis
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3. Semi-Supervised Learning

Definition: Uses both labeled and unlabeled data

When to Use:

- Large unlabeled dataset
- Labeling is expensive/time-consuming
- Small amount of labeled data available

Approaches:

- **Self-training:** Use model predictions as pseudo-labels
- **Co-training:** Multiple models trained on different views

- **Graph-based methods:** Propagate labels through graph
- **Generative models:** Learn data distribution

Common Techniques:

- Pseudo-labeling
- Consistency regularization
- MixMatch, FixMatch
- Label propagation

Use Cases:

- Medical imaging (few labeled scans)
 - Speech recognition
 - Web page classification
 - Protein function prediction
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4. Reinforcement Learning (RL)

Definition: Agent learns by interacting with environment, receiving rewards/penalties

Key Components:

- **Agent:** Learner/decision maker
- **Environment:** World agent interacts with
- **State:** Current situation
- **Action:** What agent can do
- **Reward:** Feedback signal
- **Policy:** Strategy for choosing actions

Types:

- **Model-Free:** Learn directly from experience
 - Q-Learning, SARSA
 - DQN (Deep Q-Network)
 - Policy Gradient (REINFORCE, PPO, A3C)
- **Model-Based:** Learn model of environment
 - Dyna-Q
 - MBPO (Model-Based Policy Optimization)
 - MuZero

Approaches:

- **Value-based:** Learn value function (Q-Learning, DQN)
- **Policy-based:** Learn policy directly (Policy Gradient, PPO)
- **Actor-Critic:** Combine both (A3C, SAC, PPO)

Use Cases:

- Game playing (Chess, Go, Atari, Dota)
 - Robotics and control
 - Autonomous vehicles
 - Resource management
 - Recommendation systems
 - LLM alignment (ChatGPT RLHF)
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5. Self-Supervised Learning

Definition: Creates supervision signal from data itself (no manual labels)

How it Works:

- Generate labels automatically from data structure
- Predict part of input from other parts
- Learn representations through pretext tasks

Common Techniques:

For NLP:

- **Masked Language Modeling:** Predict masked words (BERT)
- **Next Sentence Prediction:** Predict if sentences follow each other
- **Causal Language Modeling:** Predict next word (GPT)

For Computer Vision:

- **Contrastive Learning:** Similar images close, different images far
 - SimCLR, MoCo, BYOL
- **Rotation prediction:** Predict image rotation angle
- **Jigsaw puzzles:** Solve shuffled image patches
- **Colorization:** Predict color from grayscale
- **Inpainting:** Fill in missing image regions

Use Cases:

- Pre-training large models (BERT, GPT, ResNet)
 - Learning from unlabeled images/text
 - Transfer learning
 - Feature extraction
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6. Transfer Learning

Definition: Using knowledge from one task to improve learning on another task

Approaches:

- **Feature Extraction:** Use pre-trained model as fixed feature extractor
- **Fine-tuning:** Retrain some/all layers on new task
- **Domain Adaptation:** Adapt model to new domain

Common Pre-trained Models:**Computer Vision:**

- ResNet, VGG, Inception
- EfficientNet, Vision Transformers (ViT)

NLP:

- BERT, GPT, T5, RoBERTa
- LLaMA, Mistral

When to Use:

- Limited training data
- Similar tasks
- Expensive to train from scratch

Use Cases:

- Medical imaging (use ImageNet pre-trained models)
 - Custom text classification (use BERT)
 - Object detection (use COCO pre-trained models)
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7. Multi-Task Learning

Definition: Training single model on multiple related tasks simultaneously

Benefits:

- Shared representations across tasks
- Improved generalization
- Better performance on all tasks
- Efficient use of parameters

Approaches:

- **Hard parameter sharing:** Shared hidden layers
- **Soft parameter sharing:** Regularize to encourage similarity
- **Task-specific heads:** Shared backbone, separate outputs

Use Cases:

- Face recognition + age/gender prediction
 - Multi-lingual translation
 - Self-driving (detection + segmentation + depth)
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8. Active Learning

Definition: Model selects most informative samples for labeling

Query Strategies:

- **Uncertainty Sampling:** Label samples with highest uncertainty
- **Query by Committee:** Multiple models vote, label disagreements
- **Expected Model Change:** Label samples causing biggest model change
- **Diversity Sampling:** Ensure diverse sample selection

When to Use:

- Labeling is expensive (medical diagnosis, legal documents)
- Limited labeling budget
- Large unlabeled dataset

Use Cases:

- Medical image annotation
 - Document classification
 - Spam detection
 - Sentiment analysis
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9. Federated Learning

Definition: Train model across decentralized devices without sharing data

How it Works:

1. Server sends model to devices
2. Devices train locally on their data
3. Devices send only model updates to server
4. Server aggregates updates
5. Repeat

Benefits:

- **Privacy:** Data never leaves device
- **Bandwidth:** Send model updates, not raw data
- **Distributed data:** Learn from data across locations

Challenges:

- Non-IID data (different distributions)
- Communication costs
- System heterogeneity
- Security (malicious updates)

Use Cases:

- Mobile keyboard prediction (Gboard)
- Healthcare (learn from hospitals without sharing patient data)
- IoT devices
- Financial institutions

10. Meta-Learning (Learning to Learn)

Definition: Learning algorithm that learns to learn new tasks quickly

Goal: Quickly adapt to new tasks with few examples (few-shot learning)

Approaches:

- **MAML** (Model-Agnostic Meta-Learning): Learn good initialization
- **Prototypical Networks**: Learn embedding space
- **Matching Networks**: Attention-based matching
- **Memory-Augmented Networks**: External memory

Use Cases:

- Few-shot classification (recognize new classes from few examples)
 - Robot adaptation to new tasks
 - Personalization with limited user data
 - Drug discovery
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11. Online Learning

Definition: Model learns incrementally as new data arrives

Types:

- **Stochastic Gradient Descent**: Update after each sample
- **Mini-batch online**: Update after small batches
- **Streaming**: Continuous data stream

Challenges:

- Concept drift (data distribution changes)
- Catastrophic forgetting
- Memory constraints

Use Cases:

- Stock trading (real-time market data)
 - Fraud detection (evolving fraud patterns)
 - Recommendation systems (user preferences change)
 - Web search ranking
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12. Ensemble Learning

Definition: Combining multiple models for better predictions

Types:

Bagging (Bootstrap Aggregating):

- Train models on random subsets (with replacement)
- Aggregate predictions (vote/average)
- Reduces variance
- Example: Random Forest

Boosting:

- Train models sequentially, each correcting previous errors
- Weighted combination
- Reduces bias
- Examples: AdaBoost, Gradient Boosting, XGBoost

Stacking:

- Train multiple models (base learners)
- Meta-model learns to combine their predictions
- Can use different model types

Voting:

- Hard voting: Majority vote
- Soft voting: Average probabilities

Use Cases:

- Kaggle competitions (winners often use ensembles)
 - Production systems (combine multiple models)
 - Critical applications (ensemble for robustness)
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13. Imitation Learning (Behavioral Cloning)

Definition: Learn policy by imitating expert demonstrations

Types:

- **Behavioral Cloning:** Supervised learning from expert actions
- **Inverse Reinforcement Learning:** Learn reward function from expert
- **Apprenticeship Learning:** Learn to match expert's feature expectations

Challenges:

- Compounding errors
- Distribution shift
- Needs expert demonstrations

Use Cases:

- Autonomous driving (learn from human drivers)
 - Robotics (learn manipulation from demonstrations)
 - Game AI (learn from expert players)
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Comparison Summary

Type	Labeled Data	Use Case	Example
Supervised	☑ All	Classification, Regression	Spam detection
Unsupervised	✗ None	Clustering, Dimensionality reduction	Customer segmentation
Semi-Supervised	⚠ Some	When labeling is expensive	Medical imaging
Reinforcement	🎯 Rewards	Sequential decisions	Game playing
Self-Supervised	🔄 Auto-generated	Pre-training large models	BERT, GPT
Transfer	♻ Pre-trained	Limited data, similar tasks	Fine-tune BERT
Multi-Task	☑ Multiple tasks	Related tasks	Face analysis
Active	🎯 Selectively query	Expensive labeling	Medical annotation
Federated	📱 Distributed	Privacy-sensitive	Mobile keyboards
Meta	🚀 Many tasks	Few-shot learning	Quick adaptation
Online	📊 Streaming	Continuous data	Fraud detection
Ensemble	🤝 Combine models	Improve performance	Random Forest
Imitation	👤 Expert demos	Learn from experts	Autonomous driving

Fundamentals (Questions 1-10)

Q1. What is the difference between supervised and unsupervised learning?

Answer:

- **Supervised Learning:** Uses labeled data (input-output pairs) to train models. Examples: Classification, Regression
- **Unsupervised Learning:** Uses unlabeled data to find patterns. Examples: Clustering, Dimensionality Reduction

Q2. What is overfitting and how can you prevent it?

Answer: Overfitting occurs when a model learns training data too well, including noise, performing poorly on new data. **Prevention methods:**

- Cross-validation
- Regularization (L1, L2)
- Dropout (neural networks)
- Early stopping

- More training data
- Reduce model complexity

Q3. Explain the bias-variance tradeoff.

Answer:

- **Bias:** Error from incorrect assumptions. High bias = underfitting
- **Variance:** Error from sensitivity to training data. High variance = overfitting
- **Tradeoff:** Reducing one often increases the other. Goal is to find optimal balance.

Q4. What is cross-validation and why is it important?

Answer: Cross-validation splits data into multiple folds, training on some and testing on others, rotating through all combinations. **Importance:**

- Better performance estimate
- Reduces overfitting
- Makes efficient use of limited data
- K-fold CV is most common (k=5 or 10)

Q5. What is the difference between L1 and L2 regularization?

Answer:

- **L1 (Lasso):** Adds absolute value of weights. Can zero out features (feature selection). Formula: $\lambda \sum |w|$
- **L2 (Ridge):** Adds squared weights. Shrinks but doesn't eliminate features. Formula: $\lambda \sum w^2$
- **Elastic Net:** Combines both L1 and L2

Q6. Explain precision, recall, and F1-score.

Answer:

- **Precision:** $TP/(TP+FP)$ - Of predicted positives, how many are correct?
- **Recall:** $TP/(TP+FN)$ - Of actual positives, how many did we find?
- **F1-Score:** $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$ - Harmonic mean of both

Q7. What is the ROC curve and AUC?

Answer:

- **ROC Curve:** Plots True Positive Rate vs False Positive Rate at various thresholds
- **AUC:** Area Under the Curve. Ranges from 0-1
 - 1.0 = Perfect classifier
 - 0.5 = Random classifier
 - <0.5 = Worse than random

Q8. What is gradient descent and its variants?

Answer: Optimization algorithm to minimize loss function. **Variants:**

- **Batch GD:** Uses entire dataset per update (slow but stable)

- **Stochastic GD (SGD):** Uses single sample (fast but noisy)
- **Mini-batch GD:** Uses small batches (best of both)
- **Adam:** Adaptive learning rate (most popular)
- **RMSprop:** Divides learning rate by running average of gradients

Q9. What is the curse of dimensionality?

Answer: As dimensions increase:

- Data becomes sparse
- Distance metrics become less meaningful
- More data needed exponentially
- Models become more complex **Solutions:** PCA, Feature selection, Regularization

Q10. What is feature scaling and why is it important?

Answer: Transforms features to similar scales. **Methods:**

- **Normalization:** $(x - \min) / (\max - \min) \rightarrow [0, 1]$
- **Standardization:** $(x - \mu) / \sigma \rightarrow \text{mean}=0, \text{std}=1$ **Importance:**
- Required for distance-based algorithms (KNN, SVM)
- Speeds up gradient descent
- Prevents feature domination

Algorithms (Questions 11-25)

Q11. Explain decision trees and their advantages/disadvantages.

Answer: Tree-like model making decisions based on feature values. **Advantages:** Interpretable, handles non-linear, no scaling needed **Disadvantages:** Overfits easily, unstable, greedy algorithm **Improvements:** Random Forest, Gradient Boosting

Q12. What is Random Forest and how does it work?

Answer: Ensemble of decision trees using:

- **Bagging:** Bootstrap sampling of data
- **Random feature selection:** Each split uses random subset of features
- **Voting:** Classification by majority vote, regression by average **Advantages:** Reduces overfitting, handles large datasets, feature importance

Q13. Explain Gradient Boosting and XGBoost.

Answer: **Gradient Boosting:** Sequentially builds trees, each correcting previous errors. **XGBoost:** Optimized implementation with:

- Regularization (L1+L2)
- Parallel processing
- Tree pruning
- Built-in cross-validation

- Handles missing values

Q14. What is K-Means clustering?

Answer: Partitions data into K clusters. **Algorithm:**

1. Initialize K centroids randomly
 2. Assign points to nearest centroid
 3. Recalculate centroids
 4. Repeat until convergence
- Limitations:** Must specify K, sensitive to initialization, assumes spherical clusters

Q15. Explain Support Vector Machines (SVM).

Answer: Finds optimal hyperplane to separate classes. **Key concepts:**

- **Margin:** Distance between hyperplane and nearest points
- **Support Vectors:** Points closest to hyperplane
- **Kernel Trick:** Maps to higher dimensions (linear, RBF, polynomial) **Use:** Binary classification, works well with high dimensions

Q16. What is Principal Component Analysis (PCA)?

Answer: Dimensionality reduction finding principal components (directions of max variance). **Steps:**

1. Standardize data
 2. Compute covariance matrix
 3. Find eigenvectors/eigenvalues
 4. Select top K components
- Use:** Visualization, noise reduction, feature extraction

Q17. Explain K-Nearest Neighbors (KNN).

Answer: Classification/regression based on K nearest neighbors. **Process:**

1. Calculate distance to all points
 2. Find K nearest neighbors
 3. Majority vote (classification) or average (regression)
- Considerations:**
- Choose K (odd for classification)
 - Distance metric (Euclidean, Manhattan)
 - Computationally expensive at prediction

Q18. What is Naive Bayes and when to use it?

Answer: Probabilistic classifier using Bayes' theorem assuming feature independence. **Formula:** $P(y|X) = \frac{P(X|y) \times P(y)}{P(X)}$ **Types:** Gaussian, Multinomial, Bernoulli **Use cases:** Text classification, spam detection **Pros:** Fast, works with small data, handles high dimensions

Q19. Explain linear and logistic regression differences.

Answer: Linear Regression:

- Predicts continuous values
- Output: Real number
- Loss: MSE

Logistic Regression:

- Predicts probabilities (classification)
- Output: 0-1 via sigmoid function
- Loss: Binary cross-entropy

Q20. What is ensemble learning?

Answer: Combines multiple models for better performance. **Types:**

- **Bagging:** Parallel models, reduce variance (Random Forest)
- **Boosting:** Sequential models, reduce bias (XGBoost, AdaBoost)
- **Stacking:** Different model types, meta-learner combines predictions

Q21. What is the difference between batch, mini-batch, and online learning?

Answer:

- **Batch:** Train on entire dataset at once (offline learning)
- **Mini-batch:** Train on small batches (most common)
- **Online:** Train on one sample at a time, updates incrementally (streaming data)

Q22. Explain feature engineering and its importance.

Answer: Creating new features from existing data. **Techniques:**

- Polynomial features
- Interactions
- Binning/discretization
- Date features (day, month, hour)
- Text features (TF-IDF, embeddings)
- Domain-specific features **Impact:** Often more important than algorithm choice

Q23. What is a confusion matrix?

Answer: Table showing classification results.

		Predicted	
Actual	Pos	TP	FN
	Neg	FP	TN

Metrics derived: Accuracy, Precision, Recall, F1, Specificity

Q24. What is the difference between parametric and non-parametric models?

Answer: Parametric: Fixed number of parameters (Linear Regression, Logistic Regression)

- Faster, less data needed
- Strong assumptions about data

Non-parametric: Parameters grow with data (KNN, Decision Trees)

- More flexible
- Needs more data

Q25. Explain bagging vs boosting.

Answer: Bagging:

- Parallel training
- Reduces variance
- Equal weight to all models
- Example: Random Forest

Boosting:

- Sequential training
- Reduces bias
- Weights based on performance
- Example: XGBoost, AdaBoost

Deep Learning Basics (Questions 26-35)

Q26. What is a neural network?

Answer: Network of interconnected neurons (nodes) organized in layers. **Components:**

- **Input layer:** Receives features
- **Hidden layers:** Process information
- **Output layer:** Produces predictions
- **Weights:** Connection strengths
- **Activation functions:** Add non-linearity

Q27. Explain common activation functions.

Answer:

- **Sigmoid:** $\sigma(x) = 1/(1+e^{-x}) \rightarrow [0,1]$ (binary classification output)
- **Tanh:** $\tanh(x) \rightarrow [-1,1]$ (hidden layers)
- **ReLU:** $\max(0,x)$ (most popular, solves vanishing gradient)
- **Leaky ReLU:** $\max(0.01x, x)$ (prevents dying ReLU)
- **Softmax:** Multi-class classification output

Q28. What is backpropagation?

Answer: Algorithm to compute gradients for neural network training. **Process:**

1. Forward pass: Compute predictions
2. Calculate loss
3. Backward pass: Compute gradients using chain rule
4. Update weights using gradient descent

Q29. What is the vanishing gradient problem?

Answer: Gradients become very small in deep networks, slowing learning in early layers. **Causes:** Sigmoid/tanh activations, deep networks **Solutions:**

- ReLU activation
- Batch normalization
- Residual connections (ResNet)
- Better initialization (Xavier, He)

Q30. Explain batch normalization.

Answer: Normalizes layer inputs during training. **Benefits:**

- Faster training
- Higher learning rates possible
- Reduces internal covariate shift
- Acts as regularization **Formula:** $(x - \mu) / \sqrt{(\sigma^2 + \epsilon)}$

Q31. What is dropout and why use it?

Answer: Randomly drops neurons during training. **How:** Each neuron has probability p of being dropped **Benefits:**

- Prevents overfitting
- Forces network to learn redundant representations
- Ensemble effect **Typical:** 0.2-0.5 dropout rate

Q32. Explain convolutional neural networks (CNN).

Answer: Specialized for grid-like data (images). **Layers:**

- **Convolution:** Applies filters to extract features
- **Pooling:** Reduces spatial dimensions (max, average)
- **Fully connected:** Final classification **Use:** Image classification, object detection, segmentation

Q33. What is transfer learning?

Answer: Using pre-trained model on new task. **Approaches:**

- **Feature extraction:** Freeze early layers, train final layers
- **Fine-tuning:** Unfreeze and train all/some layers **Benefits:** Less data needed, faster training, better performance **Popular models:** ResNet, VGG, BERT, GPT

Q34. Explain learning rate and its importance.

Answer: Step size for weight updates.

- **Too high:** Training unstable, may not converge
- **Too low:** Very slow training, may get stuck **Strategies:**
- Learning rate decay
- Adaptive learning (Adam)
- Cyclical learning rates
- Learning rate warmup

Q35. What is data augmentation?

Answer: Creating new training samples from existing data. **Images:** Rotation, flip, crop, color jitter, mixup

Text: Synonym replacement, back-translation **Benefits:** More training data, reduces overfitting, improves generalization

Model Evaluation (Questions 36-42)

Q36. What is stratified sampling and when to use it?

Answer: Sampling that preserves class distribution. **Use:**

- Imbalanced datasets
- Cross-validation
- Train-test split **Ensures:** Each fold/split has representative class distribution

Q37. Explain mean squared error (MSE) vs mean absolute error (MAE).

Answer: MSE: Average of squared errors

- Penalizes large errors more
- Differentiable everywhere
- Sensitive to outliers

MAE: Average of absolute errors

- Linear penalty
- More robust to outliers
- Used when outliers are expected

Q38. What is class imbalance and how to handle it?

Answer: When classes have very different frequencies. **Solutions:**

- **Resampling:** Oversample minority, undersample majority
- **SMOTE:** Synthetic minority oversampling
- **Class weights:** Penalize misclassifications of minority class more
- **Anomaly detection:** If extremely imbalanced
- **Metrics:** Use F1, AUC instead of accuracy

Q39. Explain A/B testing in ML context.

Answer: Comparing two model versions in production. **Process:**

1. Split traffic (50/50 or other ratio)
2. Measure key metrics
3. Statistical significance test
4. Choose winner **Considerations:** Sample size, duration, statistical power

Q40. What is model calibration?

Answer: Ensuring predicted probabilities match actual frequencies. **Methods:**

- Platt scaling
- Isotonic regression
- Temperature scaling **Check:** Calibration plot, Brier score

Q41. Explain the concept of statistical significance.

Answer: Likelihood results are not due to chance. **Key concepts:**

- **P-value:** Probability of observing results if null hypothesis true
- **Significance level (α):** Threshold (typically 0.05)
- **Confidence interval:** Range containing true value with X% confidence **Use:** Validating model improvements, A/B tests

Q42. What is model drift and concept drift?

Answer: Model drift: Model performance degrades over time **Concept drift:** Statistical properties of target variable change **Types:**

- Sudden drift: Abrupt change
- Gradual drift: Slow change
- Recurring drift: Cyclical patterns **Detection:** Monitor performance metrics, distribution shifts

Advanced Topics (Questions 43-50)

Q43. Explain the difference between online and offline learning.

Answer: Offline (Batch):

- Train on fixed dataset
- Retrain periodically
- More stable

Online:

- Continuous learning from new data
- Adapts to changes
- Requires monitoring for drift

Q44. What is semi-supervised learning?

Answer: Uses both labeled and unlabeled data. **Approaches:**

- Self-training: Use model predictions as labels
- Co-training: Multiple views of data
- Graph-based methods **Use**: When labeling is expensive, large unlabeled data available

Q45. Explain active learning.

Answer: Model selects most informative samples for labeling. **Strategies**:

- Uncertainty sampling: Label samples model is most uncertain about
- Query by committee: Multiple models vote
- Expected model change **Benefit**: Reduces labeling cost

Q46. What is multi-task learning?

Answer: Training one model on multiple related tasks simultaneously. **Benefits**:

- Shared representations
- Regularization effect
- Better generalization **Example**: Face recognition + age estimation + gender classification

Q47. Explain federated learning.

Answer: Training model across decentralized devices without sharing data. **Process**:

1. Send model to devices
2. Train locally on each device
3. Send only model updates to server
4. Aggregate updates **Benefits**: Privacy preservation, works with distributed data

Q48. What is meta-learning (learning to learn)?

Answer: Training models to quickly adapt to new tasks. **Approaches**:

- MAML (Model-Agnostic Meta-Learning)
- Prototypical networks
- Metric learning **Use**: Few-shot learning, rapid adaptation

Q49. Explain AutoML.

Answer: Automated machine learning pipeline. **Automates**:

- Feature engineering
- Model selection
- Hyperparameter tuning
- Architecture search (NAS) **Tools**: Auto-sklearn, TPOT, H2O AutoML, Google AutoML

Q50. What is explainable AI (XAI)?

Answer: Making model predictions interpretable. **Techniques**:

- **SHAP**: Shapley values for feature importance
- **LIME**: Local interpretable model-agnostic explanations

- **Attention visualization:** For neural networks
- **Partial dependence plots**
- **Feature importance:** Tree-based models **Importance:** Trust, debugging, compliance, fairness

Reinforcement Learning (Questions 51-65)

Q51. What is Reinforcement Learning and how does it differ from supervised learning?

Answer: Reinforcement Learning (RL): Agent learns by interacting with environment, receiving rewards/penalties.

Key differences from Supervised Learning:

- **No labeled data:** Learns from trial and error
- **Delayed rewards:** Actions have long-term consequences
- **Sequential decisions:** Current action affects future states
- **Exploration vs exploitation:** Must balance trying new actions vs using known good ones

Example: Game playing (Chess, Go), robotics, recommendation systems

Q52. Explain the key components of RL: Agent, Environment, State, Action, Reward.

Answer:

Agent: The learner/decision maker **Environment:** Everything agent interacts with **State (s):** Current situation of the agent **Action (a):** Choices agent can make **Reward (r):** Feedback signal (scalar value)

MDP (Markov Decision Process):

- States: S
- Actions: A
- Transition function: $P(s'|s,a)$
- Reward function: $R(s,a,s')$
- Discount factor: $\gamma \in [0,1]$

Goal: Maximize cumulative reward (return)

Q53. What is the difference between value-based and policy-based methods?

Answer:

Value-Based Methods:

- Learn value function $V(s)$ or $Q(s,a)$
- Derive policy from values
- Examples: Q-Learning, DQN
- Good for: Discrete action spaces
- Formula: $\pi(s) = \operatorname{argmax}_a Q(s,a)$

Policy-Based Methods:

- Directly learn policy $\pi(a|s)$
- No explicit value function
- Examples: REINFORCE, PPO
- Good for: Continuous actions, stochastic policies
- Formula: $\pi_{\theta}(a|s)$ parameterized by θ

Actor-Critic: Combines both (learns policy + value function)

Q54. Explain Q-Learning algorithm.

Answer: Model-free, off-policy, value-based algorithm.

Q-Table Update Rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Where:

- α : Learning rate
- γ : Discount factor
- r : Immediate reward
- s' : Next state
- $\max_{a'} Q(s', a')$: Best future value

Algorithm:

1. Initialize $Q(s, a)$ arbitrarily
2. For each episode:
 - Choose action using ϵ -greedy policy
 - Take action, observe r, s'
 - Update $Q(s, a)$
 - $s \leftarrow s'$

Limitations: Only works for discrete, small state/action spaces

Q55. What is Deep Q-Network (DQN) and why was it revolutionary?

Answer: Uses deep neural network to approximate Q-function for large state spaces.

Key Innovations:

1. **Experience Replay:** Store transitions (s, a, r, s') in replay buffer, sample randomly
 - Breaks correlation between consecutive samples
 - Improves data efficiency
 - More stable learning
2. **Target Network:** Separate network for calculating target Q-values

- Reduces oscillations
- Updated every N steps
- Formula: $y = r + \gamma \max_{a'} Q_{\text{target}}(s', a')$

3. Loss Function:

$$L = E[(r + \gamma \max_{a'} Q_{\text{target}}(s', a') - Q(s, a))^2]$$

Breakthrough: Solved Atari games from raw pixels (2015)

Limitations:

- Overestimation of Q-values
- Cannot handle continuous actions
- Sample inefficient

Q56. Explain exploration vs exploitation tradeoff.

Answer: Fundamental dilemma in RL.

Exploration: Try new actions to discover potentially better options **Exploitation:** Use current best-known action to maximize reward

Strategies:

1. ϵ -Greedy:

- Exploit: Choose best action (1- ϵ)
- Explore: Random action (ϵ)
- Decay ϵ over time

2. Softmax/Boltzmann:

$$P(a|s) = \exp(Q(s, a)/\tau) / \sum_{a'} \exp(Q(s, a')/\tau)$$

- τ (temperature) controls randomness

3. Upper Confidence Bound (UCB):

- Choose actions with high uncertainty
- Optimistic in face of uncertainty

4. Thompson Sampling:

- Bayesian approach
- Sample from posterior distribution

In practice: Start with high exploration, decay to exploitation

Q57. What is the difference between on-policy and off-policy learning?

Answer:

On-Policy:

- Learns policy being followed
- Updates based on actions from current policy
- Examples: SARSA, A3C, PPO
- More stable, but less sample efficient

SARSA Update:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma Q(s',a') - Q(s,a)]$$

(Uses actual next action a')

Off-Policy:

- Learns optimal policy while following different (behavior) policy
- Can learn from old experiences
- Examples: Q-Learning, DQN, DDPG
- More sample efficient, but can be unstable

Q-Learning Update:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

(Uses max over all actions)

Trade-off: Sample efficiency vs stability

Q58. Explain Policy Gradient methods.

Answer: Directly optimize policy parameters to maximize expected return.

Objective:

$$J(\theta) = E_{\pi}[\sum_t \gamma^t r_t]$$

Policy Gradient Theorem:

$$\nabla_{\theta} J(\theta) = E_{\pi}[\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi}(s,a)]$$

REINFORCE Algorithm:

1. Generate episode using π_θ
2. For each step t :
 - Calculate return $G_t = \sum_{k=t}^T \gamma^{k-t} r_k$
 - Update: $\theta \leftarrow \theta + \alpha \nabla_\theta \log \pi_\theta(a_t|s_t) G_t$

Advantages:

- Works with continuous actions
- Can learn stochastic policies
- Better convergence guarantees

Disadvantages:

- High variance
- Sample inefficient
- Requires full episodes

Improvements: Actor-Critic, PPO, TRPO

Q59. What is Actor-Critic method?

Answer: Combines policy-based (actor) and value-based (critic) methods.

Components:

Actor: Policy network $\pi_\theta(a|s)$

- Selects actions
- Updated using policy gradient

Critic: Value network $V_\phi(s)$ or $Q_\phi(s,a)$

- Evaluates actions
- Reduces variance of policy gradient

Update Rules:

Critic Update (TD error):

$$\begin{aligned}\delta &= r + \gamma V(s') - V(s) \\ \phi &\leftarrow \phi + \alpha_{\text{critic}} \delta \nabla_\phi V_\phi(s)\end{aligned}$$

Actor Update:

$$\theta \leftarrow \theta + \alpha_{\text{actor}} \delta \nabla_\theta \log \pi_\theta(a|s)$$

Advantages:

- Lower variance than pure policy gradient
- More sample efficient
- Works with continuous actions

Popular Variants:

- A3C (Asynchronous Advantage Actor-Critic)
- A2C (Synchronous A3C)
- PPO (Proximal Policy Optimization)
- SAC (Soft Actor-Critic)

Q60. Explain PPO (Proximal Policy Optimization).

Answer: State-of-the-art policy gradient method with stability guarantees.

Problem with vanilla policy gradient: Large policy updates can be destructive

PPO Solution: Clip policy updates to stay close to old policy

Clipped Objective:

$$L^{\text{CLIP}}(\theta) = E[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)]$$

where:

$r_t(\theta) = \pi_{\theta}(a_t|s_t) / \pi_{\theta_{\text{old}}}(a_t|s_t)$ (probability ratio)

$\hat{A}_t$ = Advantage estimate

ϵ = clipping parameter (typically 0.2)

Intuition:

- If advantage > 0 (good action): allow increase but clip at $1+\epsilon$
- If advantage < 0 (bad action): allow decrease but clip at $1-\epsilon$

Why PPO is popular:

- Simple to implement
- Stable training
- Sample efficient
- Works well in practice
- Used by OpenAI for ChatGPT RLHF

Hyperparameters:

- Clip range: 0.1-0.3
- Epochs per update: 3-10
- Mini-batch size: 32-512

Q61. What is the discount factor (γ) and how does it affect learning?

Answer: Determines importance of future rewards.

Return (cumulative reward):

$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots = \sum_{k=0}^{\infty} \gamma^k r_{t+k}$$

γ values:

- $\gamma = 0$: Myopic (only immediate reward)
- $\gamma = 1$: Far-sighted (all future rewards equally)
- $\gamma \in (0,1)$: Balance (typical: 0.9-0.99)

Effects:

High γ (e.g., 0.99):

- Long-term planning
- Slower learning (many steps to propagate)
- Good for: Games with long episodes

Low γ (e.g., 0.9):

- Short-term focus
- Faster learning
- Good for: Quick decision tasks

Choosing γ :

- Episode length matters
- Domain-specific
- Start with 0.95-0.99
- Tune as hyperparameter

Q62. Explain model-based vs model-free RL.

Answer:

Model-Free RL:

- Learns directly from experience
- No model of environment dynamics
- Examples: Q-Learning, DQN, PPO
- **Pros:** No need to learn dynamics, works when model unknown
- **Cons:** Sample inefficient, requires many interactions

Model-Based RL:

- Learns model of environment: $P(s'|s,a)$ and $R(s,a)$

- Uses model for planning
- Examples: Dyna-Q, MBPO, MuZero
- **Pros:** Sample efficient, can simulate experience
- **Cons:** Model errors compound, harder to implement

Hybrid Approaches:

- **Dyna:** Model-free learning + model-based planning
- **MuZero:** Learned model + Monte Carlo Tree Search
- **World Models:** Learn latent dynamics model

When to use:

- Model-free: Unknown dynamics, simple enough to learn directly
- Model-based: Sample-expensive environments (robotics), known dynamics

Q63. What is reward shaping and why is it important?

Answer: Modifying reward function to guide learning without changing optimal policy.

Problem: Sparse rewards make learning very slow

- Example: Robot reaching goal only gets reward at end

Reward Shaping: Add potential-based shaping:

$$R'(s, a, s') = R(s, a, s') + \gamma \Phi(s') - \Phi(s)$$

Where $\Phi(s)$ is potential function

Examples:

1. **Distance-based shaping** (robot navigation):

$$\Phi(s) = -\text{distance_to_goal}(s)$$

2. **Progress rewards** (game):

- Checkpoint rewards
- Subgoal rewards

3. **Curriculum learning:**

- Start with easier tasks
- Gradually increase difficulty

Caution:

- Bad shaping can lead to suboptimal policy
 - Potential-based shaping preserves optimal policy
 - Don't reward proxy behaviors too much
-

Q64. Explain multi-agent RL and its challenges.

Answer: Multiple agents learning simultaneously in shared environment.

Types:

1. **Cooperative:** Agents share common goal
 - Example: Multi-robot coordination
 - Methods: Centralized training, decentralized execution (CTDE)
2. **Competitive:** Agents have opposing goals
 - Example: Game playing (poker, Dota)
 - Methods: Self-play, Nash equilibrium
3. **Mixed:** Both cooperation and competition
 - Example: Autonomous driving

Challenges:

1. **Non-stationarity:**
 - Environment changes as other agents learn
 - Violates Markov property
2. **Credit assignment:**
 - Which agent contributed to reward?
 - Solutions: QMIX, COMA
3. **Scalability:**
 - Joint action space grows exponentially
 - Communication overhead
4. **Convergence:**
 - No guarantee of convergence
 - May oscillate or diverge

Solutions:

- **QMIX:** Factorized Q-functions
 - **MADDPG:** Multi-agent DDPG
 - **Communication protocols:** Learn to communicate
 - **Mean field approximation:** Handle large number of agents
-

Q65. What is RLHF (Reinforcement Learning from Human Feedback)?

Answer: Uses human preferences to train reward model, then optimize policy with RL.

Process (used in ChatGPT, Claude):

1. Supervised Fine-tuning (SFT):

- Train on human demonstrations
- Creates baseline policy

2. Reward Modeling:

- Collect human comparisons (A better than B)
- Train reward model to predict preferences
- Loss: Bradley-Terry model

3. RL Optimization:

- Use PPO to maximize learned reward
- Add KL penalty to stay close to SFT policy
- Objective: $\text{reward} - \beta * \text{KL}(\pi || \pi_{\text{SFT}})$

Why RLHF:

- Hard to specify good reward function manually
- Humans can judge quality but can't provide scalar rewards
- Aligns AI with human preferences

Challenges:

- Expensive (requires human labelers)
- Reward model can be gamed ("reward hacking")
- Human preferences can be inconsistent
- Distribution shift from training

Variants:

- **DPO (Direct Preference Optimization):** Skips reward modeling step
- **RLAIF:** Uses AI feedback instead of human

Applications:

- Large language models (ChatGPT, Claude)
- Image generation (Stable Diffusion)
- Robotics alignment

Additional RL Concepts (Quick Reference)

Q66. What is Temporal Difference (TD) Learning?

Answer: Learns from incomplete episodes using bootstrapping.

TD(0) Update:

$$V(s) \leftarrow V(s) + \alpha[r + \gamma V(s') - V(s)]$$

Advantages over Monte Carlo:

- Online learning (update each step)
- Works with continuing tasks
- Lower variance

TD(λ): Balances between TD(0) and Monte Carlo using eligibility traces

Q67. What is importance sampling in RL?

Answer: Technique to estimate expectations under one distribution using samples from another.

Used in off-policy learning:

$$E_{\pi}[f] = E_b[\rho f]$$

where $\rho = \pi(a|s) / b(a|s)$ (importance ratio)

Problem: High variance when distributions differ **Solutions:**

- Weighted importance sampling
 - Clip importance ratios
 - Use on-policy methods (PPO)
-

Q68. What is Monte Carlo Tree Search (MCTS)?

Answer: Planning algorithm that builds search tree through simulation.

Steps:

1. **Selection:** Traverse tree using UCB
2. **Expansion:** Add new node
3. **Simulation:** Rollout to terminal state
4. **Backpropagation:** Update values

Used in: AlphaGo, AlphaZero, MuZero

Q69. What are common RL benchmarks?

Answer:

Classic Control:

- CartPole, MountainCar, Pendulum
- Simple, quick to train

Atari Games:

- 57 games from pixels
- Standard DQN benchmark

MuJoCo:

- Physics simulations
- Continuous control (Humanoid, Ant)

Robotics:

- Robot manipulation
- Fetch, Shadow Hand

Board Games:

- Chess, Go, Poker
-

Q70. What are key challenges in applying RL to real-world problems?

Answer:**1. Sample Efficiency:**

- Real-world interactions expensive
- Solutions: Sim-to-real, model-based RL

2. Safety:

- Cannot explore dangerous states
- Solutions: Safe RL, constrained RL

3. Partial Observability:

- Cannot observe full state
- Solutions: POMDPs, recurrent policies

4. Reward Specification:

- Hard to define good reward
- Solutions: Inverse RL, RLHF

5. Generalization:

- Must work in new situations
- Solutions: Domain randomization, meta-RL

6. Deployment:

- Distribution shift
- Monitoring required
- Online adaptation

Quick Tips for Interview

1. **Always clarify:** Ask about data size, constraints, use case
2. **Trade-offs:** Every method has pros/cons - mention both
3. **Practical:** Relate to real-world applications
4. **Metrics:** Know which metrics to use when
5. **Start simple:** Begin with simple baseline, then improve
6. **Validation:** Always mention cross-validation and testing
7. **Production:** Think about deployment, monitoring, maintenance

Good luck with your interview! 🚀